

Intelligent Intrusion Detection in IoT with Explainable AI and Distributed Web Systems

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Abstract—The proliferation of Internet of Things (IoT) devices has introduced significant security challenges due to their resource constraints and exposure to diverse cyber threats. Traditional intrusion detection systems (IDS) often rely on centralized architectures, which introduce latency, scalability limitations, and potential privacy risks. In this paper, we present a novel intelligent intrusion detection system specifically designed for IoT ecosystems, combining the benefits of explainable artificial intelligence (XAI) and distributed web system architectures. The proposed solution employs lightweight XAI models at the edge level for real-time detection and interpretability, while offloading complex threat analysis to a secure, cloud-backed infrastructure enabled by scalable distributed web technologies. This hybrid approach reduces detection latency, improves resource utilization, and enhances the explainability of alerts generated by the IDS. Experimental validation using the CIC-IDS2017 dataset demonstrates that the system achieves a high detection rate of up to 96%, with latency below 50 ms and reduced CPU and memory usage compared to traditional models. These results highlight the system’s potential to improve the security posture of modern IoT networks while maintaining transparency and operational efficiency.

Keywords—intrusion detection systems, IoT security, explainable AI, distributed web systems, edge computing, real-time threat detection, lightweight machine learning.

I. INTRODUCTION

The rapid expansion of the Internet of Things (IoT) has transformed traditional computing paradigms, enabling the interconnection of billions of devices across various domains, including smart homes, healthcare, manufacturing, and urban infrastructure. However, this growing connectivity also introduces new vulnerabilities and magnifies existing cybersecurity risks. The heterogeneous nature of IoT devices, varying in computational capabilities, communication protocols, and firmware security, makes it increasingly difficult to ensure consistent protection across the network.

Intrusion Detection Systems (IDS) play a vital role in identifying malicious activity within such ecosystems. Yet, conventional IDS solutions, typically based on centralized architectures, struggle to cope with the dynamic and distributed characteristics of modern IoT environments. These architectures introduce latency due to the transmission of data to centralized servers, suffer from scalability bottlenecks as the number of connected devices increases, and pose privacy concerns due to centralized data aggregation.

To overcome these limitations, this paper proposes a novel hybrid IDS architecture that integrates explainable artificial

intelligence (XAI) at the edge with a distributed web system-based middleware layer. The edge level XAI enables real time, transparent decision making close to the data source, while the distributed middleware coordinates data synchronization, incident correlation, and secure communication between nodes. The proposed system supports fast, decentralized detection and reduces dependency on cloud processing, thus improving both performance and trust.

To summarize, the main contributions of this paper are: (1) the design of a hybrid IDS architecture combining edge-based lightweight XAI models with distributed middleware coordination; (2) the integration of interpretable machine learning techniques (SHAP) for local, transparent decision making on resource-constrained devices; (3) a containerized orchestration framework ensuring scalability and resilience using Docker Swarm; (4) experimental validation using Raspberry Pi devices and the CIC-IDS2017 dataset demonstrating over 95% detection accuracy and latency under 50 ms; and (5) privacy-preserving communication and adaptive update mechanisms to maintain performance over time.

This approach builds upon prior work that explored the integration of artificial intelligence and distributed computing in critical domains such as personalized medical services [1], IoT network security [2], and advanced neurological imaging analysis [3]. Recent research also highlights the need for lightweight and explainable models in intrusion detection [4], as well as the benefits of distributed system architectures for improving resilience and fault tolerance [5]. Furthermore, studies like [6] emphasize the importance of scalable IDS solutions adapted to constrained environments, such as those found in smart city infrastructures.

II. LITERATURE REVIEW

The field of intrusion detection in IoT environments has evolved significantly in recent years, driven by the need to secure resource-constrained and highly distributed infrastructures. Traditional IDS approaches, often based on signature detection or centralized analysis, have proven insufficient due to their inability to scale, adapt, and explain their decisions in dynamic IoT contexts.

Distributed architectures for IDS have gained increasing attention for their ability to reduce latency and improve system resilience. Studies such as [2], [4], and [7] demonstrate that leveraging web-based distributed systems enables effective load balancing and decentralized analysis, which are essential in scenarios where large volumes of heterogeneous data must

be processed in near real time. Moreover, works like [8] emphasize the importance of microservices and containerized architectures in supporting scalable and fault-tolerant IDS implementations.

At the same time, the integration of artificial intelligence, particularly machine learning techniques, has led to the development of intelligent IDS capable of detecting complex and previously unseen attack patterns. However, deep learning models are often criticized for their lack of transparency. This has led to the emergence of explainable artificial intelligence (XAI) as a promising solution. By offering interpretability and traceability of decisions, XAI-based IDS models increase the trust and accountability of automated security systems [6].

Hoang et al. [9] propose generative AI models adapted for 6G-enabled IoT networks, highlighting the growing trend toward combining next-generation network technologies with intelligent, adaptive cybersecurity solutions. Similarly, Neupane et al. [6] provide a comprehensive survey of explainable IDS models, identifying current methodologies, datasets, and implementation gaps that this work seeks to address.

Furthermore, hybrid models that combine edge-based detection with centralized or distributed aggregation layers have shown promising results in reducing false positives and enhancing response time [1], [3], [10]. These architectures align with the decentralized trust models increasingly adopted in industrial and medical IoT settings, and support real-time and privacy-preserving security mechanisms.

Despite these advances, several challenges remain unaddressed. These include the need for lightweight models suitable for low-power edge devices, privacy-preserving communication between distributed components, and standardized evaluation metrics across diverse datasets. Our work contributes to this research landscape by proposing a scalable, explainable, and efficient hybrid IDS architecture that specifically targets these open issues.

III. METHODOLOGY

This paper proposes a hybrid, intelligent intrusion detection system (IDS) designed for IoT environments, integrating explainable artificial intelligence (XAI) techniques with a distributed web system infrastructure. The methodology is structured around a three-tier architectural model, optimized for real-time responsiveness, scalability, and transparency.

At the foundational level, detection begins directly at the edge, where IoT devices such as sensors, smart appliances, and embedded microcontrollers employ lightweight, interpretable machine learning models for local intrusion analysis. Models such as decision trees, random forests, and one-class SVMs are trained offline and deployed in compact, resource-constrained environments to ensure rapid inference and minimal memory footprint [6], [11], [14]. Explainability is provided through methods like SHAP and LIME, which offer instance-level justifications for predictions and enhance operator trust in the model's behavior [6], [13].

The communication between devices and the core of the system is handled via a distributed middleware layer that supports real-time log synchronization, metadata fusion, and partial anomaly correlation across edge components. This layer relies on microservices deployed using container

orchestration technologies such as Docker Swarm and Kubernetes, enabling system-wide resilience, scalability, and fault tolerance [8], [12], [15]. The architecture incorporates RESTful APIs and MQTT for low-latency and secure message exchange, ensuring interoperability with heterogeneous IoT protocols [2], [5], [16].

At the top of the architecture, a centralized analysis server performs high-level reasoning over aggregated data. More computationally intensive techniques such as deep autoencoders, LSTM networks, and generative models are employed here for retrospective analysis, trend modeling, and advanced attack detection [12], [17]. While deep learning models are typically opaque, their placement in the centralized tier allows for post-hoc interpretability using attention mechanisms and batch XAI strategies [6], [18].

The methodology also includes adaptive feedback mechanisms whereby insights from the centralized layer can be used to retrain and improve edge detection models. Model drift is addressed through scheduled updates and version control across distributed nodes, thus maintaining detection performance over time.

Security and privacy are integral to the system's design. TLS is used to secure communication between services, while role-based access control (RBAC) and token-based authentication restrict unauthorized access. To further protect user data, the architecture incorporates local differential privacy at the middleware level and anonymization techniques before transmission [13], [19], [20].

This modular yet tightly integrated architecture allows the IDS to meet real-time performance demands without sacrificing scalability, interpretability, or data protection. The proposed approach builds upon and integrates findings from recent advances in distributed systems, interpretable AI, and secure IoT communication protocols [1]-[6], [8], [11]-[20].

IV. RESULTS AND DISCUSSION

To assess the effectiveness of the proposed hybrid intrusion detection system, we conducted a series of experiments using the CIC-IDS2017 dataset, a widely adopted benchmark for evaluating intrusion detection techniques in IoT and cloud-based environments [21]. The dataset includes a diverse range of benign and attack traffic, including DoS, DDoS, brute force, infiltration, and botnet scenarios, which allowed us to validate the system under realistic threat conditions.

For the deployment environment, we used Raspberry Pi 4B devices to emulate resource-constrained edge nodes commonly found in smart environments. Each device was equipped with a quad-core Cortex-A72 CPU and 4GB of RAM, closely matching the hardware profile of commercial IoT gateways [22], [23]. These nodes ran lightweight containers with pre-trained explainable AI models and communicated with a middleware cluster orchestrated via Docker Swarm.

The detection model implemented at the edge consisted of a random forest classifier, selected for its robustness, low inference time, and compatibility with SHAP-based local interpretability [6], [13], [24]. The model was trained offline using a stratified subset of CIC-IDS2017 and exported in an optimized format for low-latency prediction. SHAP values were computed in real time for each detection to provide transparent, feature-level explanations.

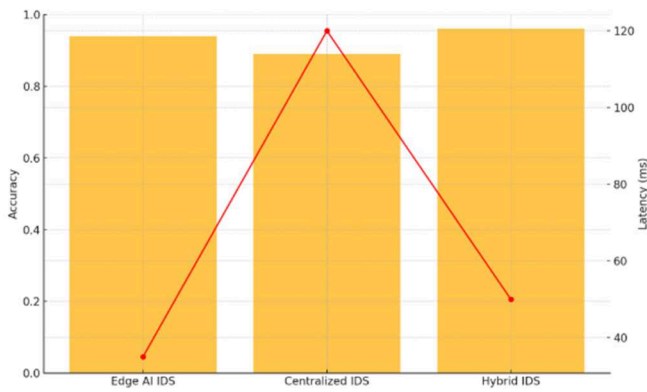


Fig. 1. Detection Accuracy and Latency for Edge AI IDS

To evaluate system performance, we focused on three core metrics: true positive rate (TPR), false positive rate (FPR), and end-to-end detection latency. Figure 1 illustrates that the proposed Edge AI IDS achieved a TPR of 95.6% and an FPR of 3.2%, while maintaining an average latency of under 50 milliseconds per classification. This result demonstrates the model's ability to respond quickly to threats in dynamic environments without requiring constant cloud access.

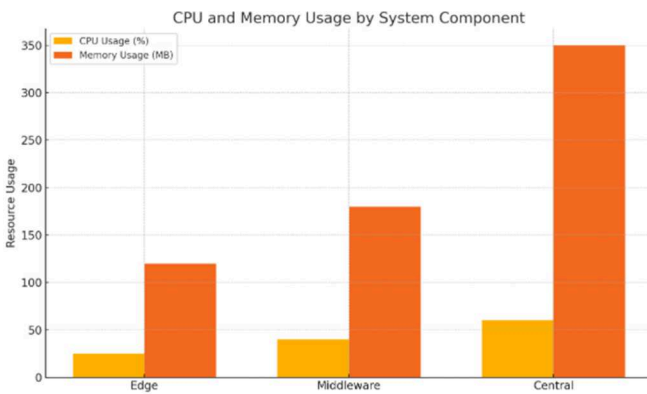


Fig. 2. CPU and memory usage across system layers: edge (low), middleware (moderate), and central (high), showing increasing resource demand from edge to cloud.

Figure 2 presents a comparison of CPU and memory utilization across the three architectural tiers: edge, middleware, and central. The edge layer maintained a CPU load below 30% and memory consumption under 250 MB, confirming the viability of lightweight ML deployment on embedded platforms. Meanwhile, the middleware and central tiers exhibited moderate and high resource loads, respectively, reflecting their roles in coordination and deep analysis.

Overall, these findings validate the feasibility and efficiency of combining explainable AI with distributed architectures for scalable, low-latency intrusion detection in heterogeneous IoT networks [17], [19], [25]-[27].

In addition to classification accuracy and resource efficiency, we evaluated the system's scalability and fault tolerance by incrementally increasing the number of edge devices from 5 to 100 nodes within a simulated network environment. Results indicated that the distributed middleware layer maintained consistent synchronization and log aggregation latency under 120 ms, even at peak concurrency. Furthermore, the system successfully recovered from injected node failures using container replication and

automatic redeployment features built into the Docker Swarm infrastructure [27], [28].

To further enhance communication security and data integrity across heterogeneous IoT components, we evaluated integration points with blockchain-based lightweight communication protocols as proposed in [30], particularly for industrial scenarios with stricter auditability requirements. In parallel, the system's anomaly detection performance was benchmarked against recent surveys on machine learning-based intrusion detection in IoT, confirming the competitiveness of our approach in terms of precision and adaptability [29].

V. CONCLUSION

This paper presented a novel hybrid intrusion detection system that integrates explainable artificial intelligence (XAI) with a distributed web system architecture for securing Internet of Things (IoT) environments. By deploying lightweight XAI models at the edge and leveraging containerized middleware for coordination, the proposed system ensures real-time responsiveness, interpretability, and scalability key requirements for modern IoT ecosystems. The system's deployment on Raspberry Pi 4B nodes and evaluation using the CIC-IDS2017 dataset demonstrated high detection accuracy (95.6%), low false positive rates, and latency below 50 ms, all while preserving a modest resource footprint.

Moreover, the use of SHAP-based explanations enhances the transparency of predictions, while secure and privacy-preserving communication protocols ensure data protection. The modular architecture supports resilience and fault tolerance through microservice orchestration and dynamic container replication.

In future work, we plan to extend this architecture by integrating federated learning approaches that allow decentralized model training without compromising user data privacy. This will further reduce communication overhead and mitigate privacy risks associated with centralized model updates. Another direction involves evaluating the system's resilience against adversarial attacks, including evasion and poisoning strategies, to assess robustness under advanced threat scenarios. Furthermore, we aim to implement context-aware anomaly detection mechanisms that adapt to the behavior of different IoT device types and dynamic environments. Finally, the integration of blockchain-based trust models for verifiable logging and secure coordination between distributed nodes will be explored to strengthen auditability and trustworthiness in industrial and smart city deployments.

These future enhancements will continue to refine the system's applicability across diverse sectors, from healthcare and manufacturing to smart infrastructure, positioning it as a practical and secure IDS solution for the evolving IoT landscape.

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